

Rangamati Science and Technology University

Department of Computer Science and Engineering
4th year 2nd semester B. Sc. (Engg.) Final Exam-2023

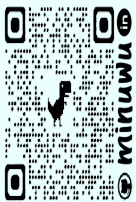
Course code: CSE 4202

Course Title: Engineering Management

Time: 1 Hours

Full Marks: 15

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1. Management has been defined as “getting things done through people”. Comment on this definition. 3
 2. Briefly discuss on the managerial functions. 3
 3. Why has Frederick Taylor been called “the father of scientific management”? 3
 4. What is the relationship between planning and controlling? 3
 5. Briefly discuss on the scope of engineering ~~economics~~ *management* in light of the “Universality of Management” concept. 3



Rangamati Science and Technology University

Department of Computer Science and Engineering

4th Year 2nd Semester B.Sc. (Engg.) Mid-term Exam-2023

Course Title: Digital System Design Course Code: CSE-4203

Session: 2019-20

Time: 1 Hours

Marks: 15

NB: 1. Answer all questions. 2. Figures in the right margin indicate marks.

1.	Design a 4-bit arithmetic logic circuit that performs the addition, subtraction, increment, decrement and transfer A operations on two 4-bit inputs $A = A_3A_2A_1A_0$ and $B = B_3B_2B_1B_0$, selectable by a 3-bit opcode. For the design you must: i. List the opcode table ii. Show the bit-level logical expressions used to realize add and subtract in a single ripple adder stage. Explain how subtraction is implemented with the adder. iii. Briefly explain the operation for Increment and Decrement (how control lines and B/carry are arranged), and mention what status flags you would provide (Carry, Overflow, Zero) and how to get them.	10
2	Using Register Transfer Language (RTL), write the sequence of micro-operations required to execute each of the following operations in a simple processor. Also explain the role of the control signals in each step. a) Load data from memory location M into register R1. b) Add the contents of R1 and R2 and store the result in R3. c) Increment the Program Counter (PC) after fetching an instruction.	05

10 + 5 = 15



Rangamati Science and Technology University

Department of Computer Science and Engineering

4th Year 2nd Semester, Class Test-1 B.Sc. Examination-2023

Course Title: Information Security

Course Code: CSE-4205

Time: 50 minute

Marks: 10

[Answer the following questions. Figures in the right-hand margin indicate full marks.]

1. What do mean by Information system? Describe the classification of Information system in details. 4
2. What are the Critical characteristics of Information Security? Write the Limitations of Mobile Computing. 3
3. What kind of security **Threats & Attacks** that can affect mobile devices has face due to base of security? Explain it. 3



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4th Year 2nd Semester, Mid Term-1 B.Sc. Examination-2024

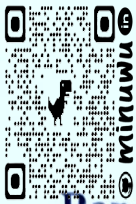
Course Title: Information Security **Course Code: CSE-4205**

Time: 1 Hour

Marks: 15

[Answer the following questions. Figures in the right-hand margin indicate full marks.]

1. (a) What kind of methods is used by Intrusion prevention systems to detect security incidents? Describe those in details. 4
- (b) Distinguish between: 4
 - a. False Negative & False Positive
 - b. Synchronous Token & Asynchronous Token
2. (a) Derive the process for creating a session key between two parties for communication using Needham-Schroeder protocol. 3
3. Explain the Approach process of Fixed password-based authentication for authentication. 4



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4th Year 2nd Semester Mid-01 Examination-2023

Course Title: Machine Learning

Time: 01 Hour Marks: 15

Course Code: CSE-5210

1. a. Define Machine Learning with example. 1
- b. Considering following path graph and graph of pheromone using Ant Colony Optimization algorithm we get probability of selecting a path taking $\alpha \neq \beta \neq 1$. 4

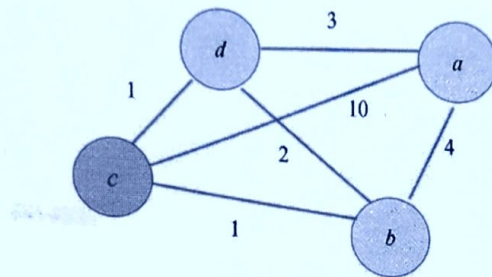


Fig a: Length of path

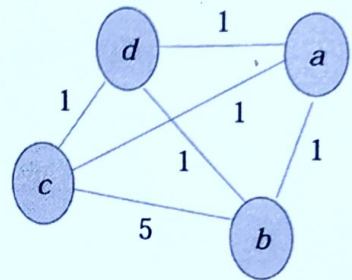


Fig b: Concentration of pheromone

2. a. Shortly describe the several techniques of selection of population for mating pool in machine learning evolutionary algorithms. 2
- b. State the application of swarm intelligence. Draw the flow chart of particle swarm optimization (PSO) algorithm. 3
3. a. Define employed bees and Scout bees. 1
- b. Consider an objective function, $f(x) = x^5 - 11x^4 + 27x^3 - 49x^2 + 30x$ and the initial values of five food sources are given in table below. Apply ABC (Artificial bee colony) to find minimum value of the function (complete only employed bees and onlooker bees phase). 4

Food	2.5608	0.8128	2.8992	0.1582	1.9672
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